Classification of Positive Natural Collatz Seeds with Mechanical Parity Itineraries

Collatz proof project

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Abstract

We complete the repository's mechanical-word branch. For arbitrary real slope and intercept, a positive natural seed with a globally mechanical shortcut Collatz parity itinerary must be 1 or 2, and its slope must be 1/2. Every positive orbit with a mechanical tail therefore reaches 1. The proof combines a factor-complexity bound, exact rational phase coding, and the endpoint-swap cycle obstruction. All real intercepts and nonprimitive rational presentations are handled. No universal assertion that Collatz itineraries become mechanical is proved. This is a formalization and integration of classical ingredients, not a proof of Collatz or a claim of mathematical novelty.

1 Statements

Let $T(n) = n/2$ for even n and $(3n+1)/2$ for odd n. The predicate $\mathrm{HasItin}(\alpha, \rho, N)$ means that, at every natural time t,

$$T^t(N) \bmod 2 = \lfloor (t+1)\alpha + \rho \rfloor - \lfloor t\alpha + \rho \rfloor.$$

Theorem 1. *For all $\alpha, \rho \in \mathbb{R}$ and $N \in \mathbb{N}$ with $N > 0$, $\mathrm{HasItin}(\alpha, \rho, N)$ implies $\alpha = 1/2$ and $N \in \{1, 2\}$.*

Theorem 2. *If $N > 0$ and $\mathrm{HasItin}(\alpha, \rho, T^s(N))$ for some natural s and real α, ρ, then $T^t(N) = 1$ for some natural t.*

The positivity condition is essential: zero has the all-zero itinerary. The first theorem restricts seeds and slope; it does not assert that every intercept realizes both seeds.

2 Periods and rotations are actual returns

If $q\alpha$ is an integer, translating the two floors by this integer shows that the mechanical word is q-periodic. Naturals with identical infinite parity itineraries are equal, by congruence modulo 2^L for every L. Hence $T^q(N) = N$ for any natural seed with this itinerary. In particular, a rational denominator is a return period from the initial seed, without a boundedness assumption.

A realized return word rotated by s is realized by $T^s(N)$. This is proved by reducing indices modulo the period and pumping the return. The endpoint-swap theorem therefore applies to rotations: if a return word has both $1u0$ and $0u1$ as rotations, then u is empty and the corresponding shifted seeds are 1 and 2.

3 A uniform finite rational grid

For $0 < p < q$ and $\gcd(p, q) = 1$, define the carry bit

$$b_r(t) = \mathbf{1}\{(tp + r) \bmod q + p \geq q\}.$$

Multiplication by p permutes residues modulo q. Thus, from any phase r, some shifts reach phase 0 and phase $q-1$. For $0 < i < q$, the residue $ip \bmod q$ is nonzero. This gives the following structure for the two extreme length-q words:

$$(b_{q-1}(0), \ldots, b_{q-1}(q-1)) = 1u0, \qquad (b_0(0), \ldots, b_0(q-1)) = 0u1.$$

Indeed, the first letters are 1 and 0 and the last letters are 0 and 1. At interior positions, replacing the residue by one less cannot change the carry: equality at the threshold would make $(i+1)p$ divisible by q. The grid is periodic, so both shifted words are actual return words. Endpoint cancellation now forces a visit to 1.

4 All real intercepts and all rational presentations

Normalize ρ to $\beta = \{\rho\}$ and put $r = \lfloor q\beta \rfloor$. The inequalities $r \le q\beta < r+1$, together with quotient and remainder arithmetic, prove for every natural t that

$$\left\lfloor t\frac{p}{q} + \beta \right\rfloor = \left\lfloor \frac{tp+r}{q} \right\rfloor .$$

Successive differences are exactly $b_r(t)$. This is an equality of infinite words, including negative original intercepts and threshold boundary cases, not a finite or floating-point approximation. Consequently every reduced rational mechanical itinerary in $(0,1)$ reaches 1.

For an arbitrary real slope, the earlier mechanical complexity bound $p_N(L) \le L+1$ forces boundedness. Repeated states give a positive return period q on a tail. Mechanical telescoping over k repeats gives $-1 < k(q\alpha - p) < 1$, where p is the odd-step count. Hence $q\alpha = p$. Positive-cycle inequalities give $0 < p < q$. Dividing p, q by their gcd produces a reduced rational slope, so the grid theorem applies to this tail and the original seed reaches 1. No rational presentation is assumed primitive in this step.

Finally shift to the visit to 1. Its two-step return and telescoping force $2\alpha = 1$. This integral slope increment gives a two-step return at the original seed. The checked two-step-return classification gives $N = 1$ or 2.

5 Formal artifact and validation

The new modules are `MechanicalPeriod.lean`, `RationalGrid.lean`, and `RationalMechanical.lean` under `lean/Collatz/`. The principal declarations are `mechanical_reaches_one`, `mechanical_classification`, and `eventual_mechanical_reaches_one` in namespace `Collatz`. The modular phase, endpoint structure, real-intercept transport, and period transport are separately proved and reusable.

The Lean build and selected axiom audit pass: 107 audited declarations use only the standard foundations. Four exact computational controls pass, including negative and boundary intercepts and the two permitted seeds. These tests are regression checks; the universal classification is a Lean proof. The audit is dependency inspection, not independent kernel replay.

6 Context and the remaining Collatz gap

The complexity ingredient is related to Dubickas, *On integer sequences generated by linear maps* (2009), Theorem 5, doi:10.1017/S0017089508004655. The cycle cancellation follows Knight's high-cycle argument, *Collatz high cycles do not exist* (2026), doi:10.1016/j.disc.2025.114812, also inspected in the nested Knight formalization. No nested module or custom mathematical axiom is imported.

The previously documented rational rotation and intercept bridges are now closed. General Collatz counterexamples need not have mechanical tails. Proving that every positive orbit eventually has such a tail would itself settle the conjecture and cannot be assumed as a coverage shortcut. General escape and nontrivial cycles remain unresolved outside this class.